Emotional Diary: Secure Personal Diary Mobile Application

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Submitted in partial fulfillment of the requirements of the degree of

Bachelor of Science in

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Declaration

We do hereby declare that the project work presented in this report entitled "Emotional Diary: Secure Personal Diary Mobile Application" is the result of our own work. We further declare that this report has been prepared and written by us and that no part of this report has been submitted elsewhere for the requirements of any degree, award, diploma, or other academic purposes.

The project has been developed as an academic software project with the objective of providing users with a secure and convenient mobile diary environment. All external resources, technologies, frameworks, libraries, and references used in the development of the system have been appropriately acknowledged.

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I do hereby declare that the project work presented in this report entitled "Emotional Diary: Secure Personal Diary Mobile Application" is the outcome of the original work carried out by 1st Group Member Name, 2nd Group Member Name, 3rd Group Member Name, 4th Group Member Name, and 5th Group Member Name under my supervision. I further declare that no part of this project report has been submitted elsewhere for the requirements of any degree, award, or diploma, or for any other academic purpose, except for publications. I further certify that this project report meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

We dedicate this project to our loving parents and family members, whose continuous support, encouragement, patience, and inspiration have been an essential part of our academic journey. Their unconditional love, guidance, and sacrifices have motivated us to overcome challenges and remain committed to our studies and project work.

We are also grateful to our teachers and mentors who have encouraged us to develop our knowledge, improve our skills, and pursue meaningful academic work. Their valuable guidance and constructive suggestions have helped us throughout the development of this project.

Finally, we dedicate this work to everyone who has supported, encouraged, and inspired us directly or indirectly during the completion of our project. We hope that the knowledge and experience gained through this work will contribute to our future academic and professional endeavors.

Acknowledgement

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Finally, we would like to thank all our group members for their cooperation, teamwork, and dedication. Every member contributed to different stages of planning, design, implementation, testing, debugging, and documentation of the project.

Abstract

Emotional Diary is a secure personal diary mobile application designed to provide users with a convenient digital environment for recording, managing, and reflecting on their daily experiences and emotions. Traditional diary applications mainly focus on storing written entries, while the proposed application attempts to provide a more meaningful and personalized diary experience by incorporating emotion analysis into diary management. The application is developed using Android Studio and Java, with Firebase Authentication used for user authentication and Firebase Firestore used for cloud-based diary data management. The system provides essential features such as account creation, secure login, diary creation, diary viewing, editing, deletion, diary searching, profile management, password management, settings, and light/dark theme support.

A major feature of the application is its AI-based emotion detection system. After a user writes a diary entry, the system analyzes the overall meaning, context, tone, and emotional characteristics of the entry and assigns one dominant emotion from a predefined set of emotions: Happy, Excited, Calm, Neutral, Sad, and Angry.

The system also considers the user's previous diary context as supporting information so that the emotion analysis can better understand the user's writing style and emotional patterns without blindly copying previous emotions.

The application also provides an emotion summary through a visual pie chart, allowing users to understand the distribution of their recorded emotions. The combination of secure diary management, personalized emotion analysis, and an intuitive mobile interface makes Emotional Diary a practical and interactive personal journaling solution.

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List of Abbreviations and Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
CSE	Computer Science and Engineering
DB	Database
GUI	Graphical User Interface
IDE	Integrated Development Environment
UI	User Interface
UX	User Experience
XML	Extensible Markup Language
SDK	Software Development Kit
CRUD	Create, Read, Update, Delete
Firebase	Google Firebase Platform
Firestore	Cloud Firestore Database

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Chapter 1

1 Introduction

Digital technology has significantly changed the way people store, communicate, and manage personal information. Personal diary applications have become an alternative to traditional paper diaries because they provide greater convenience, accessibility, searchability, and storage capability. However, many conventional diary applications primarily focus on storing written information and provide limited assistance in understanding the emotional characteristics of diary entries.

Emotional Diary has been developed to address this limitation by combining personal diary management with artificial intelligence-based emotion analysis. The application allows users to create personal diary entries and automatically determine the dominant emotion associated with each entry. Instead of relying only on individual keywords, the system considers the overall meaning and context of the newly written diary.

The application also considers previous diary information as supporting context. This allows the emotion analysis system to understand the user's personal writing patterns while ensuring that the newly written diary remains the strongest source of evidence for emotion classification.

1.1 Problem Statement

Many people record their daily experiences, thoughts, and emotions in personal diaries. Traditional paper diaries can be difficult to organize, search, protect, and access from multiple locations. Digital diary applications solve some of these problems, but many basic diary systems simply store text without providing meaningful analysis of the user's emotional state.

Another challenge is that emotions expressed in written text can be context-dependent. A diary containing words such as "good", "bad", "excited", or "sad" cannot always be accurately classified by simple keyword matching. The overall meaning, situation, tone, and relationship between different parts of the text are important.

Therefore, there is a need for a personal diary application that combines secure diary management with intelligent emotion analysis while maintaining a simple and user-friendly interface.

1.2 Problem Background

Personal journaling is often used to record daily experiences, memories, thoughts, and feelings. In a paper-based diary, users can freely write about their experiences, but the information is difficult to search and analyze. Digital diaries improve accessibility and organization, but storing diary entries alone does not necessarily provide additional insight into emotional patterns.

Modern artificial intelligence and natural language processing technologies can analyze written text and identify emotional characteristics. Such technologies provide an opportunity to enhance traditional diary systems by automatically analyzing the emotional content of diary entries.
Emotional Diary uses this concept to create a more interactive journaling environment. The application analyzes each new diary entry and classifies it into a predefined emotion category. Previous diary information may also be used as supporting context to make the analysis more personalized.

1.3 Objectives

The main objective of this project is to develop a secure, user-friendly, and intelligent personal diary mobile application that allows users to manage their diary entries and understand their emotional patterns.
The specific objectives are:

- **Secure Diary Management:** To develop a secure mobile platform for creating and managing personal diary entries.
- **User Authentication:** To implement user registration and authentication using Firebase Authentication.
- **Diary CRUD Operations:** To provide Create, Read, Update, and Delete operations for diary entries.
- **Diary Search:** To provide a search mechanism for quickly finding diary entries.
- **AI Emotion Detection:** To implement AI-based emotion detection for newly written diary entries.
- **Emotion Classification:** To classify diary entries into Happy, Excited, Calm, Neutral, Sad, and Angry.
- **Personalized Analysis:** To use previous diary context as supporting information for more personalized emotion analysis.
- **Emotion Visualization:** To provide an emotion summary through a visual pie chart.
- **Account Management:** To provide profile management, password management, and application settings.
- **Theme Support:** To support both light and dark themes for improved usability.

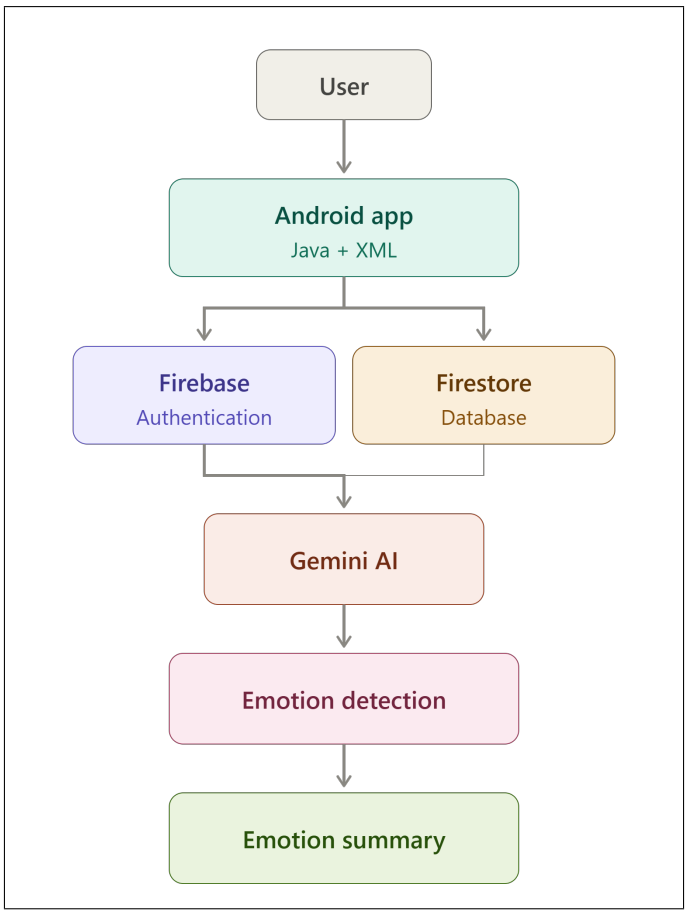


Figure 1.1: Overview of the Emotional Diary application

This System Architecture Diagram provides a comprehensive, high-level overview of the structural design and operational framework of the Emotional Diary Application. The client-side tier is built as a native Android app using Java and XML, offering a responsive user interface designed for smooth interaction, entry creation, and visual feedback.
The application leverages Google Cloud services for secure access and scalable data management. Firebase Authentication handles identity verification and session control, while Firestore Database provides real-time, NoSQL storage for diary entries, user settings, and metadata.
To deliver intelligent behavioral analysis, the system integrates with Gemini AI, which processes written entry context to execute automated Emotion detection. The extracted emotional data is then synthesized into an Emotion summary, presenting users with meaningful analytics, mood trends, and tailored psychological insights.

1.4 Motivations

The primary motivation behind this project is to improve the traditional digital diary experience by introducing intelligent emotion analysis. Writing a diary can help users record important events and personal experiences, but manually identifying emotional patterns across many diary entries can be difficult.
The project is motivated by the idea that an intelligent diary system should not only store information but also provide useful interpretation of that information. Automatic

emotion classification can help users observe the general emotional characteristics of their diary entries without requiring them to manually label every entry.
Another motivation is to create a practical academic project that combines mobile application development, cloud database management, authentication, user interface design, and artificial intelligence in a single system.

1.5 Significance of the Project

The project is significant because it combines conventional diary management with AI-assisted emotional analysis. The application provides users with a centralized environment where they can securely manage personal diary entries and review their emotional distribution.
From a technical perspective, the project demonstrates the integration of Android development with cloud-based authentication and database services. It also demonstrates how generative AI can be integrated into a mobile application for contextual text analysis.
From a usability perspective, the system provides a simple interface with search functionality, profile management, theme support, and visual emotion summaries. These features make the application more practical than a basic text storage application.

1.6 Contributions

The major contributions of this project are:

- **Complete Mobile Diary:** Development of a complete Android-based personal diary application.
- **Cloud-Based Authentication and Storage:** Implementation of Firebase-based authentication and cloud diary data storage.
- **Diary Management:** Implementation of diary creation, viewing, editing, deletion, and searching.
- **AI Emotion Detection:** Development of an AI-based emotion detection mechanism for diary entries.
- **Personalized Emotion Analysis:** Introduction of previous diary context as supporting information for personalized emotion analysis.
- **Emotion Visualization:** Development of a visual emotion summary using a pie chart.
- **Profile and Settings:** Implementation of profile management, password management, and settings.
- **Theme Support:** Implementation of professional light and dark theme support.
- **User Interface:** Development of a user-friendly mobile interface using Android components and Material Design principles.

1.7 Complex Engineering Analysis

The project involves several engineering challenges because it combines mobile software development, cloud services, authentication, database management, artificial intelligence, user interface design, and testing in a single application.

1.7.1 Complex Engineering Problems

The major complex engineering problems addressed by the project include:

- **System Architecture Design:** Designing a reliable architecture for a multi-screen Android application.
- **Data Security and Authentication:** Maintaining secure authentication and user-specific diary data.
- **Cloud Integration:** Connecting the mobile application with cloud-based Firestore data.
- **Asynchronous Data Handling:** Handling asynchronous cloud database operations.
- **AI Workflow Integration:** Integrating AI-based emotion detection into the diary creation workflow.
- **Prompt Engineering and Constraints:** Designing a prompt that ensures the AI returns only valid emotion categories.
- **Contextual History Balance:** Using previous diary information without allowing old entries to dominate the analysis of a new entry.
- **Theme and Readability Management:** Maintaining readable interfaces under both light and dark themes.

1.7.2 Complex Engineering Activities

The project required several complex engineering activities including:

- **Requirement Analysis:** Requirement analysis and feature planning.
- **UI and Navigation Design:** Designing application screens and navigation between major modules.
- **Firebase Integration:** Firebase Authentication and Firestore database integration.
- **AI Integration:** Integration of AI-based emotion detection through the Firebase AI environment.
- **Application Development:** Development and debugging of multiple Android activities and fragments.

- **Testing:** Testing authentication, diary management, emotion detection, profile management, and settings.
- **Debugging:** Debugging theme and Android resource-related problems.
- **Final UI Polishing:** Improving the visual consistency and usability of the application.

1.8 Report Organization

This report is organized into six major chapters.
Chapter 1: Introduction presents the project background, problem statement, objectives, motivation, significance, contributions, and engineering challenges.
Chapter 2: Background Study discusses related concepts, existing systems, literature review, and the problem analysis associated with intelligent personal diary applications.
Chapter 3: Methodology describes the feasibility study, requirements, technologies, system design, development model, diagrams, and overall system flow.
Chapter 4: Implementation and Testing describes the implementation of the application modules, system setup, prototype design, end-user requirements, testing, and quality assurance.
Chapter 5: Standards, Constraints, and Milestones discusses sustainability, social impact, ethics, challenges, constraints, and the project timeline.
Chapter 6: Conclusion summarizes the completed work, limitations, and possible future improvements.
In summary, this chapter introduced Emotional Diary as a secure and intelligent personal diary application. The next chapter discusses the background study and related concepts that support the development of the proposed system.

Chapter 2

2 Background Study

The development of Emotional Diary is based on several areas of computer science including mobile application development, cloud database systems, authentication, human-computer interaction, and artificial intelligence. This chapter discusses the relevant concepts and existing approaches that influenced the development of the proposed application.

2.1 Literature Review

Digital diary and journaling applications have become increasingly useful because they allow users to store personal experiences electronically. Compared with traditional paper diaries, digital systems provide advantages such as searchability, structured storage, accessibility, and easier management of diary entries.

Modern digital diary applications provide different features for recording daily experiences, emotions, memories, and personal notes. Some applications mainly focus on mood tracking, while others provide complete diary management through features such as searching, reminders, calendars, media attachment, and cloud synchronization.

Privacy and data management are also important considerations in personal diary applications because diary entries may contain sensitive and private information. Therefore, secure authentication and reliable data storage are important for protecting users' personal information.

The proposed Emotional Diary is an Android-based personal diary application developed using Java. The system provides user registration and login, diary creation, editing, viewing, deletion, and searching facilities. It uses Firebase Authentication for user account authentication and Firebase Firestore for cloud-based diary data management. The application also provides emotion recording and an emotion summary interface, allowing users to review their emotional patterns. Profile management, password management, application settings, and dark theme support are also included to provide a convenient and user-friendly experience.

A review of existing diary, journaling, mood tracking, and note-taking applications was conducted to understand their available features and limitations. A total of 25 existing applications were considered in this review. The following subsections discuss these applications and compare their major characteristics with the proposed Emotional Diary.

2.1.1 Daylio

Short Description:

Daylio [1] is a popular mood tracking and digital diary app that helps users record their daily mood and activities.

How It Works:

Users select their mood, choose activities, and add a short note. The app saves the information and tracks mood history.

Features:

- Mood tracking
- Daily journal
- Activity tracking
- Reminder

Limitations:

- Many features are premium.
- Mainly focuses on mood tracking.
- Uses cloud backup for some features.

Comparison with Emotional Diary:

Unlike Daylio, the proposed Emotional Diary focuses on complete diary management in addition to emotion recording. Users can create, edit, view, delete, and search diary entries. The application also provides user authentication, profile management, password management, and a simple emotion summary interface. Diary information is managed securely using Firebase Firestore, providing cloud-based access to stored diary entries.

2.1.2 Diaro

Short Description:

Diaro [2] is a personal diary application that helps users write and organize their daily notes and memories.

How It Works:

Users create diary entries, organize them into categories, and search previous entries easily.

Features:

- Daily journal
- Categories and tags
- Search option
- Password protection
- Cloud backup

Limitations:

- Some features require a premium subscription.
- Cloud synchronization needs an internet connection.
- The interface may seem complex for new users.

Comparison with Emotional Diary:

The proposed Emotional Diary provides the essential diary management functions through a simple and focused interface. It supports creating, editing, viewing, deleting, and searching diary entries while using Firebase Firestore for data storage. Firebase Authentication is used for user authentication, and the application also includes profile and password management facilities. These features make the system suitable for users who prefer a straightforward personal diary application.

2.1.3 Journey

Short Description:

Journey [3] is a digital diary application that helps users record their daily thoughts, memories, and emotions in one place.

How It Works:

Users create diary entries by writing text and can organize them by date. The app can also synchronize data across multiple devices.

Features:

- Daily journal
- Photo attachment
- Cloud synchronization
- Calendar view
- Reminder

Limitations:

- Many advanced features require a premium subscription.
- Cloud synchronization requires an internet connection.
- Some features may be unnecessary for users who only want a simple diary.

Comparison with Emotional Diary:

The proposed Emotional Diary is designed as a lightweight personal diary application with essential diary management features. It supports diary creation, editing, deletion, viewing, and searching with cloud data synchronization. Firebase Firestore provides cloud storage, while Firebase Authentication manages user accounts. The application also provides emotion recording, profile management, password management, and dark theme support.

2.1.4 Daybook

Short Description:

Daybook [4] is a digital diary app that allows users to record daily thoughts, emotions, and personal experiences.

How It Works:

Users write diary entries, save them by date, and can access them anytime through the application.

Features:

- Daily diary
- Mood tracking
- Reminder
- Cloud backup
- Calendar view

Limitations:

- Many features are available only in the premium version.
- Internet is required for cloud synchronization.
- Some users may find the interface feature-heavy.

Comparison with Emotional Diary:

The proposed Emotional Diary focuses on the core requirements of personal diary management. Users can manage diary entries through create, edit, view, delete, and search operations. The application uses Firebase Firestore for cloud diary storage and provides Firebase Authentication for account management. It also includes emotion recording and summary, profile management, password management, settings, and dark theme support.

2.1.5 Google Keep

Short Description:

Google Keep [5] is a note-taking application that can also be used as a simple personal diary.

How It Works:

Users create notes, lists, and reminders, which are automatically synchronized with their Google account.

Features:

- Notes
- Checklist
- Reminder
- Labels
- Cloud synchronization

Limitations:

- It is not specifically designed for diary writing.
- No dedicated mood tracking feature.
- Requires a Google account for synchronization.

Comparison with Emotional Diary:

Unlike Google Keep, the proposed Emotional Diary is specifically designed for personal diary management. It provides dedicated diary creation, editing, viewing, deletion, and searching facilities. It also includes emotion recording and an emotion summary interface, which are not the primary focus of a general note-taking application. Firebase Firestore is used for backend diary storage, providing a focused and private journaling experience.

2.1.6 Moodnotes

Short Description:

Moodnotes [6] is a mood journal application that helps users understand and improve their emotional well-being.

How It Works:

Users record their mood, write short notes, and review their emotional history.

Features:

- Mood tracking
- Daily journal
- Mood history
- Simple interface

Limitations:

- Many features are premium.
- Mainly focuses on mood tracking.
- Limited diary management features.

Comparison with Emotional Diary:

The proposed Emotional Diary combines emotion recording with complete diary management. Users can create, edit, view, delete, and search detailed diary entries instead of only recording moods. The application also provides authentication, profile management, password management, Firebase Firestore database storage, and a dark theme for improved usability.

2.1.7 MindDoc

Short Description:

MindDoc [7] is a mental health application that helps users monitor their mood and emotional condition.

How It Works:

Users answer daily questions about their feelings, and the app tracks their emotional well-being over time.

Features:

- Mood tracking
- Daily assessment
- Progress reports
- Mental health insights

Limitations:

- Mainly designed for mental health monitoring.
- Some features require a premium subscription.
- Not intended as a complete diary application.

Comparison with Emotional Diary:

The proposed Emotional Diary is not primarily a mental health assessment application. Instead, it provides a dedicated environment for recording and managing personal diary entries. Users can create, edit, delete, view, and search entries while recording their emotions. The application also provides account authentication, profile management, password management, and cloud data storage through Firebase Firestore.

2.1.8 ThinkUp

Short Description:

ThinkUp [8] is a self-improvement application that encourages positive thinking through daily affirmations.

How It Works:

Users create or listen to positive affirmations and track their personal progress.

Features:

- Positive affirmations
- Daily reminders
- Goal tracking
- Audio playback

Limitations:

- Focuses on self-improvement instead of diary writing.
- Limited journaling features.
- Many features are premium.

Comparison with Emotional Diary:

Unlike ThinkUp, the proposed Emotional Diary is specifically designed for recording personal thoughts, experiences, and emotions. It provides complete diary CRUD operations, search functionality, emotion recording, and emotion summary. In addition, Firebase Authentication, Firebase Firestore, profile management, password management, and application settings provide a complete diary management environment.

2.1.9 eMoods

Short Description:

eMoods [9] is a mood tracking application designed to help users monitor their emotional changes.

How It Works:

Users record their daily mood and symptoms, and the app keeps a history of their emotional changes.

Features:

- Mood tracking
- Daily logging
- History reports
- Reminder

Limitations:

- Mainly designed for mood monitoring.
- Limited diary writing features.
- The interface may not be suitable for users looking for a personal journal.

Comparison with Emotional Diary:

The proposed Emotional Diary provides both diary writing and emotion recording capabilities. It allows users to maintain detailed diary entries and perform create, edit, view, delete, and search operations. The emotion summary interface provides an overview of recorded emotions, while Firebase Firestore supports cloud diary storage.

2.1.10 Reflectly

Short Description:

Reflectly [10] is an AI-powered digital journal that helps users record their thoughts, feelings, and daily experiences.

How It Works:

Users write about their day, select their mood, and answer simple questions. The app organizes the entries and tracks emotional patterns.

Features:

- AI-guided journaling
- Mood tracking
- Daily reminders
- Personal journal
- Cloud backup

Limitations:

- Many features require a premium subscription.
- Internet is needed for some features.
- AI features may not suit every user.

Comparison with Emotional Diary:

The proposed Emotional Diary follows a simpler and more focused approach than AI-guided journaling applications. It does not depend on AI-based guidance for diary management. Instead, it provides direct diary creation, editing, viewing, deletion, and searching along with emotion recording and summary. Firebase Firestore database storage, Firebase Authentication, profile management, password management, and dark theme support make the application suitable for users who prefer a straightforward personal diary system.

2.1.11 Bearable

Short Description:

Bearable [11] is a health and mood tracking app that helps users record their daily emotions and activities.

How It Works:

Users log their mood and daily experiences, and the app keeps a record of their progress.

Features:

- Mood tracking
- Daily journal
- Reminder
- Progress history

Limitations:

- Mainly focuses on health tracking.
- Some features require a premium subscription.

Comparison with Emotional Diary:

The proposed Emotional Diary focuses on personal diary writing rather than health tracking. It provides complete diary management, including creating, editing, viewing, deleting, and searching entries. Emotion recording and emotion summary are integrated into the diary experience, while Firebase Firestore provides reliable cloud storage for diary information.

2.1.12 Penzu

Short Description:

Penzu [12] is an online personal diary application that allows users to keep private journal entries.

How It Works:

Users write diary entries, save them, and access them from different devices.

Features:

- Digital diary
- Password protection
- Cloud backup
- Search option

Limitations:

- Many features are premium.
- Cloud features require an internet connection.

Comparison with Emotional Diary:

The proposed Emotional Diary emphasizes secure diary management using Firebase Firestore. Users can create, edit, view, delete, and search their diary entries within the application. Firebase Authentication provides account authentication, while profile and password management facilities provide additional account control. This approach provides convenient access to synchronized diary information.

2.1.13 Grid Diary

Short Description:

Grid Diary [13] is a journaling app that helps users organize daily thoughts using guided questions.

How It Works:

Users answer daily questions and save their responses as diary entries.

Features:

- Guided journaling
- Daily diary
- Reminder
- Calendar view

Limitations:

- Guided format may not suit every user.
- Premium features are available.

Comparison with Emotional Diary:

The proposed Emotional Diary provides users with freedom to write their diary entries without requiring a fixed guided-question format. It supports complete diary management through create, edit, view, delete, and search operations. Users can also record emotions and review the emotion summary. The simple interface and dark theme provide additional flexibility for everyday use.

2.1.14 Stoic Journal

Short Description:

Stoic Journal [14] is a self-reflection app that combines journaling with mindfulness exercises.

How It Works:

Users write journal entries, answer reflection questions, and track their emotions.

Features:

- Journal writing
- Mood tracking
- Daily reflection
- Reminder

Limitations:

- Some features require a subscription.
- More focused on self-improvement than diary writing.

Comparison with Emotional Diary:

The proposed Emotional Diary focuses primarily on personal diary writing and management. Users can record their own thoughts and experiences, manage entries through CRUD operations, search previous entries, and record emotions. It also provides user authentication, profile management, password management, and Firebase Firestore cloud storage without requiring a guided self-reflection approach.

2.1.15 Gratitude

Short Description:

Gratitude [15] is a journal app that encourages users to record positive thoughts and daily gratitude.

How It Works:

Users write gratitude notes and maintain a daily journal.

Features:

- Gratitude journal
- Daily notes
- Reminder
- Mood tracking

Limitations:

- Mainly focuses on gratitude journaling.
- Advanced features require a premium version.

Comparison with Emotional Diary:

Unlike gratitude-focused applications, the proposed Emotional Diary allows users to record different types of daily experiences and emotions without restricting entries to positive thoughts. It provides complete diary management, emotion recording, searching, profile management, and password management. Firebase Firestore is used for backend diary data storage, supporting a focused personal journaling experience.

2.1.16 My Diary

Short Description:

My Diary [16] is a personal diary app that allows users to record daily memories and emotions.

How It Works:

Users write diary entries, add photos, and organize them by date.

Features:

- Personal diary
- Photo support
- Calendar view
- Password lock

Limitations:

- Some features are premium.
- Cloud synchronization requires internet.

Comparison with Emotional Diary:

The proposed Emotional Diary provides the essential functionality required for personal diary management through a simple Android interface. It supports creating, editing, viewing, deleting, and searching diary entries, together with emotion recording and summary. Firebase Authentication provides account management, while Firebase Firestore provides cloud diary storage. Profile editing, password management, settings, and dark theme support further improve usability.

2.1.17 Moodflow

Short Description:

Moodflow [17] is a mood journal that helps users record emotions and monitor mood changes.

How It Works:

Users log their mood, write short notes, and review mood history.

Features:

- Mood tracking
- Daily journal
- Statistics
- Reminder

Limitations:

- Mainly focuses on mood tracking.
- Some features require payment.

Comparison with Emotional Diary:

The proposed Emotional Diary provides broader diary management functionality in addition to emotion recording. Users can create detailed diary entries, edit them, view previous entries, delete unwanted entries, and search for specific diaries. The emotion summary interface provides a visual overview of recorded emotions. Cloud Firebase Firestore storage and Firebase Authentication support the application's data and account management requirements.

2.1.18 Presently

Short Description:

Presently [18] is a gratitude journal app designed to help users develop positive daily habits.

How It Works:

Users write daily gratitude entries and review previous records.

Features:

- Gratitude journal
- Daily reminder
- History view
- Simple interface

Limitations:

- Limited to gratitude journaling.
- Does not support complete diary management.

Comparison with Emotional Diary:

The proposed Emotional Diary is not limited to gratitude-based journaling. Users can write about different daily experiences and emotions and manage their entries through complete CRUD operations. The application also provides diary search, emotion recording, emotion summary, authentication, profile management, and password management. The use of Firebase Firestore supports cloud diary data management.

2.1.19 Journal It!

Short Description:

Journal It! [19] is a journaling application for recording daily activities, thoughts, and personal notes.

How It Works:

Users create journal entries and organize them with tags and categories.

Features:

- Daily journal
- Notes
- Tags
- Calendar

Limitations:

- Interface can be complex for new users.
- Some advanced features are premium.

Comparison with Emotional Diary:

The proposed Emotional Diary follows a simple and focused interface for managing personal diary entries. It provides create, edit, view, delete, and search functionality without requiring users to organize their entries through a complex tagging structure. Emotion recording, emotion summary, authentication, profile management, password management, and dark theme support are also included.

2.1.20 Day One Journal

Short Description:

Day One Journal [20] is a popular digital diary app for recording personal memories and daily experiences.

How It Works:

Users write journal entries, add photos, and organize them by date.

Features:

- Personal journal
- Photo attachment
- Calendar
- Cloud backup

Limitations:

- Many features require a premium subscription.
- Cloud synchronization depends on internet access.

Comparison with Emotional Diary:

The proposed Emotional Diary focuses on the essential requirements of personal diary writing and management. It provides diary CRUD operations, search functionality, emotion recording, and an emotion summary interface. Firebase Firestore is used for cloud diary data storage, while Firebase Authentication manages user accounts. The application also includes profile management, password management, settings, and dark theme support.

2.1.21 DailyBean

Short Description:

DailyBean [21] is a simple mood diary application that helps users record their daily emotions and activities.

How It Works:

Users choose their daily mood, add short notes, and save their memories in a calendar.

Features:

- Mood tracking
- Daily notes
- Calendar view
- Simple interface

Limitations:

- Limited diary writing features.
- No advanced diary management.

Comparison with Emotional Diary:

The proposed Emotional Diary provides more complete diary management functionality than a simple mood diary. Users can create, edit, view, delete, and search diary entries while also recording emotions. The application provides an emotion summary interface and supports user authentication, profile management, password management, and Firebase Firestore database storage.

2.1.22 Diary with Lock

Short Description:

Diary with Lock [22] is a private diary application that protects users' personal notes with a password.

How It Works:

Users write diary entries and access them using a password or PIN.

Features:

- Password protection
- Daily diary
- Calendar
- Search option

Limitations:

- Most advanced features require premium access.
- Limited emotion-related functionality.

Comparison with Emotional Diary:

The proposed Emotional Diary provides secure user account management through Firebase Authentication and supports complete diary operations, including creating, editing, viewing, deleting, and searching entries. In addition to diary management, the application provides emotion recording and an emotion summary interface. Profile editing, password management, settings, and dark theme support provide additional control and usability.

2.1.23 Daily Diary Journal

Short Description:

Daily Diary Journal [23] helps users record their daily experiences and personal memories.

How It Works:

Users write diary entries, save them by date, and review previous records.

Features:

- Daily journal
- Calendar
- Search
- Password lock

Limitations:

- Limited customization options.
- Limited emotion-based features.

Comparison with Emotional Diary:

The proposed Emotional Diary provides complete diary CRUD operations together with search and emotion recording. The application also provides an emotion summary interface, allowing users to review their recorded emotions. Firebase Authentication, Firebase Firestore, profile management, password management, and dark theme support provide additional functionality for a modern personal diary application.

2.1.24 Daybook Journal

Short Description:

Daybook Journal [24] is a digital diary app that allows users to record daily thoughts and personal experiences.

How It Works:

Users create diary entries, organize them by date, and manage them within the app.

Features:

- Daily diary
- Reminder
- Calendar
- Cloud backup

Limitations:

- Cloud features require internet.
- Several features are available only in the premium version.

Comparison with Emotional Diary:

The proposed Emotional Diary is designed as a simple and focused Android diary application. It supports diary creation, editing, viewing, deletion, and searching, together with emotion recording and emotion summary. Firebase Firestore database storage provides convenient management of diary information, while Firebase Authentication manages user accounts. The application also includes profile and password management and supports both light and dark themes.

2.1.25 Diary Book

Short Description:

Diary Book [25] is a personal diary application that allows users to record daily thoughts, memories, and emotions securely.

How It Works:

Users create diary entries, save them by date, and view or edit them whenever needed.

Features:

- Daily diary
- Password protection
- Calendar view
- Search option

Limitations:

- Limited customization features.
- Limited emotion-related analysis.
- Some advanced options are unavailable.

Comparison with Emotional Diary:

The proposed Emotional Diary provides a complete set of essential diary management functions, including creating, editing, viewing, deleting, and searching diary entries. It also provides emotion recording and an emotion summary interface. Firebase Authentication supports account management, while Firebase Firestore provides cloud diary data storage. Profile editing, password management, application settings, and dark theme support further improve the overall user experience.

After reviewing 25 existing diary, journaling, mood tracking, and related applications, it can be observed that existing systems provide a wide range of useful features but generally focus on specific aspects such as mood tracking, gratitude journaling, guided reflection, note taking, cloud synchronization, or general diary management. Some applications also provide premium-based features or depend heavily on cloud services.
The proposed Emotional Diary addresses these observations by combining the essential functions of a personal diary and emotion recording system in a single Android application. The system provides user registration and login, diary creation, editing, viewing, deletion, and searching facilities. It also includes emotion recording and an emotion summary interface, allowing users to review their emotional information alongside their diary entries.
From a data management perspective, Firebase Firestore is used for cloud diary storage, while Firebase Authentication is used for user account authentication. The application also provides profile editing, password management, application settings, and dark theme support. Therefore, the proposed system focuses on providing a simple, accessible, and user-friendly diary experience while maintaining reliable cloud management of diary information.

2.2 Problem Analysis

The analysis of existing personal diary applications indicates several limitations. Basic diary applications generally focus on storing text and allowing users to perform simple

operations such as creating and deleting entries. They may not provide intelligent analysis of the emotional characteristics of the entries.

Another problem is that emotion classification is not always straightforward. The same word can express different emotions depending on the context. Therefore, a simple keyword-based approach may produce inaccurate results.

Emotional Diary addresses these problems by analyzing the complete meaning of the new diary entry. The system considers the title and content together, evaluates the emotional tone and context, and selects one dominant emotion. Previous diary information is used only as supporting context.

The system also restricts the result to six predefined emotion categories. This makes the emotion summary consistent and allows the application to visualize the distribution of emotions.

In summary, the background study demonstrates the need for an application that combines conventional diary management with intelligent emotional analysis. Emotional Diary addresses this need through the integration of mobile development, cloud services, and AI-based text analysis.

Chapter 3

3 Methodology

The methodology of Emotional Diary focuses on systematic requirement analysis, feasibility evaluation, system design, implementation, and testing. The project was developed as a mobile software system using Android Studio and Java, with Firebase services used for authentication and cloud data management.

3.1 Feasibility Study

A feasibility study was conducted before implementation to determine whether the proposed application was technically, economically, and operationally practical.

Table 1: Feasibility Study Summary

Feasibility Type	**Assessment**
Economic Feasibility	The project can be developed using commonly available development tools and cloud services, reducing the requirement for expensive infrastructure.
Technical Feasibility	Android Studio, Java, Firebase Authentication, Cloud Firestore, and AI services provide the required technical capabilities.
Operational Feasibility	The application is designed for ordinary smartphone users and provides a simple interface for diary management and emotion analysis.

3.1.1 Economic Feasibility

The proposed system is economically feasible because its development mainly requires a standard computer, Android development tools, an Android smartphone or emulator, and internet connectivity. Android Studio and Java provide the main development environment without requiring specialized hardware.
Cloud services are used for authentication, database storage, and AI-based processing. The project can be developed and tested within available development resources, making it suitable for an academic software project.

3.1.2 Technical Feasibility

The project is technically feasible because all major components required for implementation are supported by established development technologies. Android Studio provides

the development environment, Java is used for application logic, Firebase Authentication manages user accounts, and Cloud Firestore manages diary information.
The AI-based emotion detection mechanism can be integrated into the application through the Firebase AI environment. The application can communicate with cloud services through supported SDKs and asynchronous operations.

3.1.3 Operational Feasibility

The system is operationally feasible because users do not require advanced technical knowledge to operate the application. The application provides simple navigation between Home, Diary, and Profile sections.
Users can create an account, log in, write diary entries, search entries, edit or delete diaries, manage their profile, change their password, and review their emotion summary through an intuitive interface.

3.2 Requirement Analysis

Requirement analysis identifies the functional and non-functional requirements necessary for the successful development of the application.

3.2.1 Functional Requirements

The major functional requirements are:

1. The system shall allow users to create an account.
2. The system shall allow registered users to log in securely.
3. The system shall support password recovery and password management.
4. The system shall allow users to create new diary entries.
5. The system shall automatically analyze the emotion of a new diary.
6. The system shall store diary entries in the user's cloud database.
7. The system shall allow users to view their previous diaries.
8. The system shall allow users to edit diary entries.
9. The system shall allow users to delete diary entries.
10. The system shall allow users to search their diary entries.
11. The system shall display an emotion summary.
12. The system shall allow users to edit profile information.
13. The system shall allow users to change their password.
14. The system shall provide application settings.
15. The system shall support light and dark themes.
16. The system shall allow users to log out securely.

3.2.2 Non-Functional Requirements

The major non-functional requirements are:

- **Security:** User authentication and user-specific data access should be properly maintained.
- **Usability:** The application should be simple and easy to understand.
- **Performance:** The application should respond efficiently during normal operations.
- **Reliability:** Diary information should be stored and retrieved consistently.
- **Maintainability:** The application should use modular activities, fragments, and components.
- **Scalability:** The cloud-based architecture should allow the system to accommodate additional users and diary records.
- **Compatibility:** The application should operate on supported Android devices.
- **Accessibility:** Text and interface elements should remain readable in both light and dark themes.

3.2.3 Tools & Technology

The main tools and technologies used in the project are:

Table 2: Tools and Technologies

Technology	**Purpose**
Android Studio	Main development environment
Java	Application programming language
XML	User interface layout development
Firebase Authentication	User registration and authentication
Cloud Firestore	Cloud-based diary data storage
Firebase AI	Integration of AI-based emotion analysis
Material Design	UI components and visual design
Git / Version Control	Source code management when required
Android Emulator / Device	Application testing

3.3 System Design

The proposed system follows a modular mobile application architecture. The main components are the user interface, authentication layer, diary management layer, cloud database, and AI-based emotion detection system.

The user interacts with the Android application through different screens. Authentication is handled through Firebase Authentication. After successful authentication, the user can access diary management features. Diary information is stored in the user's Firestore collection.
When a new diary is created, the application sends the diary information to the emotion detection component. The AI system analyzes the new diary and returns one predefined emotion. The result is then stored with the diary entry and can later be used in the emotion summary.

3.3.1 Development Model

An iterative development approach was followed for the project. The application was developed through multiple stages including requirement analysis, UI design, authentication implementation, diary management, emotion detection integration, testing, debugging, and final UI polishing.
An iterative approach was useful because several components required repeated testing and modification. For example, the dark theme required additional adjustments to background colors, text colors, input fields, and buttons to maintain readability.

3.3.2 Use Case Diagram

The use case diagram represents the major interactions between the user and the Emotional Diary system.

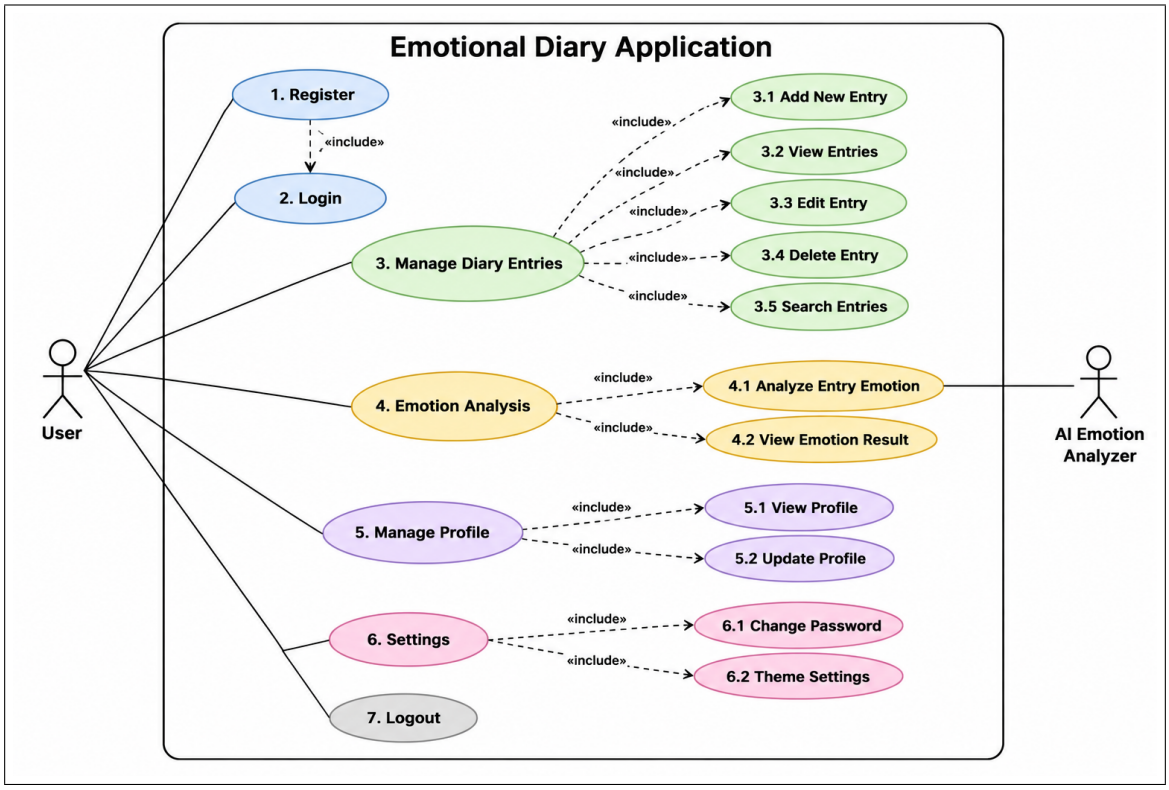


Figure 3.1: Use Case Diagram of Emotional Diary

This Use Case Diagram outlines the interaction between primary actors and the core functionalities of the Emotional Diary Application. The User interacts with primary

modules including Registration/Login, Manage Diary Entries (Add, View, Edit, Delete, Search), Manage Profile, Settings, and Logout. The AI Emotion Analyzer acts as a secondary actor that performs emotion analysis on diary entries and displays the results. Key sub-functions, such as password management and entry operations, are linked using ¡¡include¿¿ relationships to indicate mandatory sub-processes.

3.3.3 Context Diagram

The context diagram represents the interaction between the user and the overall Emotional Diary system. The user provides authentication information and diary content, while the system provides authentication results, stored diaries, emotion analysis, and emotion summaries.

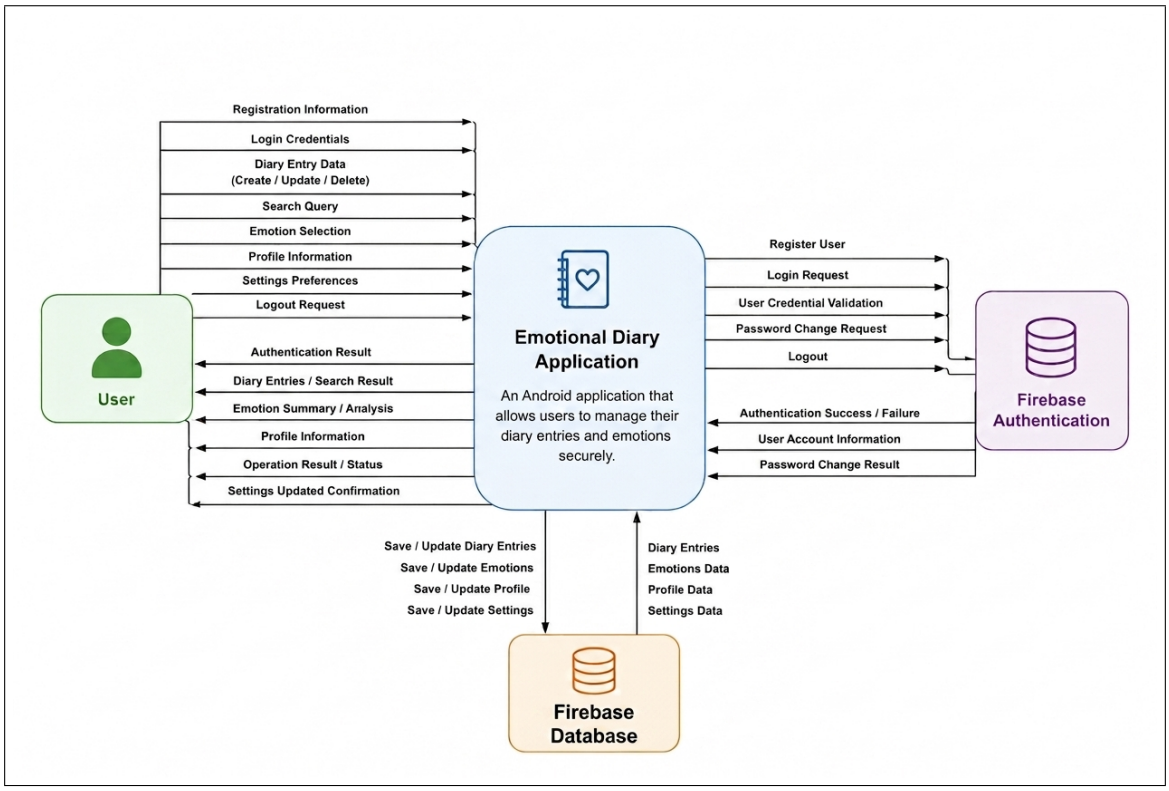


Figure 3.2: Context Diagram of Emotional Diary

This diagram illustrates the data flow between the user, the Emotional Diary Application, and the backend services. The user provides inputs such as registration, login details, diary entries, and settings updates. The application interacts with Firebase Authentication for user validation and security, while communicating with Firebase Database to save and update diary entries, emotions, and profile data in real time. Finally, the system displays authentication results, saved diary entries, emotion summaries, and status confirmations back to the user.

3.3.4 Data Flow Diagram

The Data Flow Diagram illustrates how information moves through the system. User authentication data is processed through Firebase Authentication. Diary information

is submitted to the diary management component and stored in Firestore. New diary information is also passed to the AI emotion detection component.

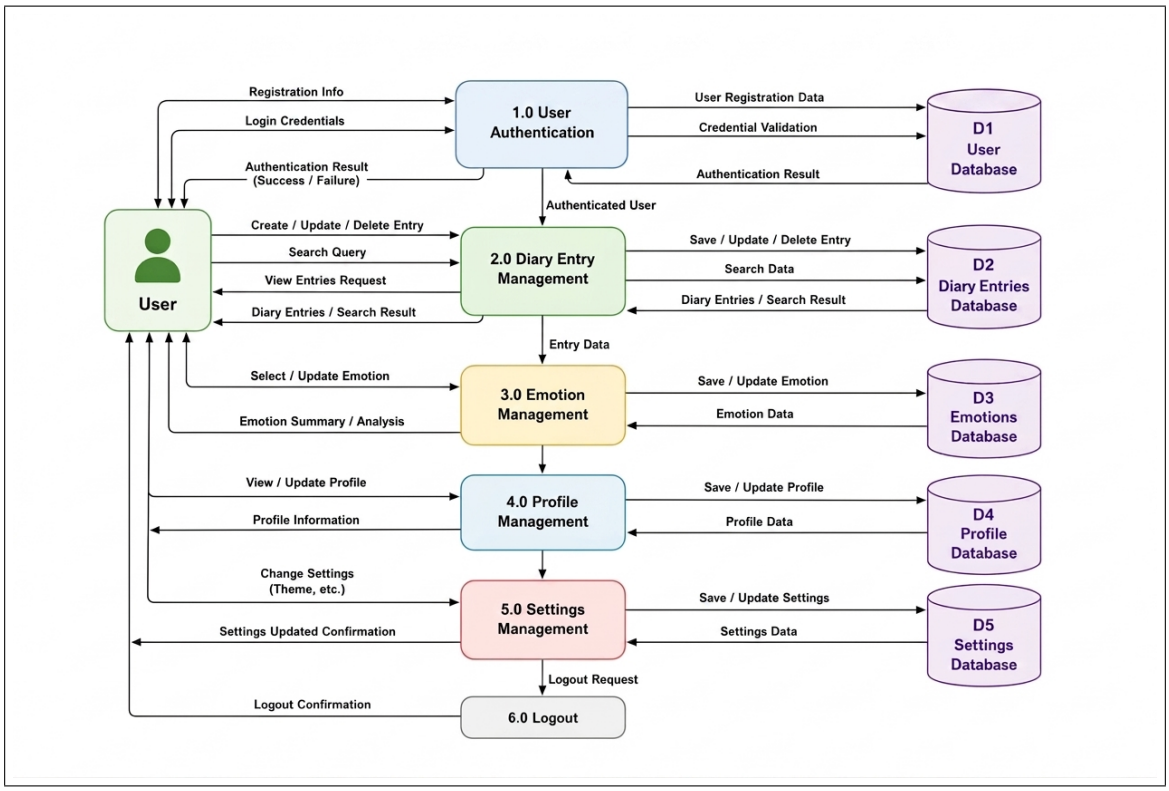


Figure 3.3: Data Flow Diagram of Emotional Diary

This Data Flow Diagram (Level 0) illustrates the detailed transformation and flow of data across the major sub-processes and the cloud database architecture. The User interacts with six main functional processes: 1.0 User Authentication, 2.0 Diary Entry Management, 3.0 Emotion Management, 4.0 Profile Management, 5.0 Settings Management, and 6.0 Logout. Each process handles specific client requests—such as authentication details, entry operations, emotion tracking, profile updates, and system preferences—and performs real-time read/write operations with Firebase Database (structured via dedicated database nodes/collections). Processed responses, including authentication status, entry search results, emotion summaries, and system confirmations, are seamlessly returned to the user.

3.3.5 Entity-Relationship Diagram

The Entity-Relationship Diagram represents the logical relationship between users and diary records. A user can have multiple diary entries, while each diary entry belongs to a specific authenticated user.

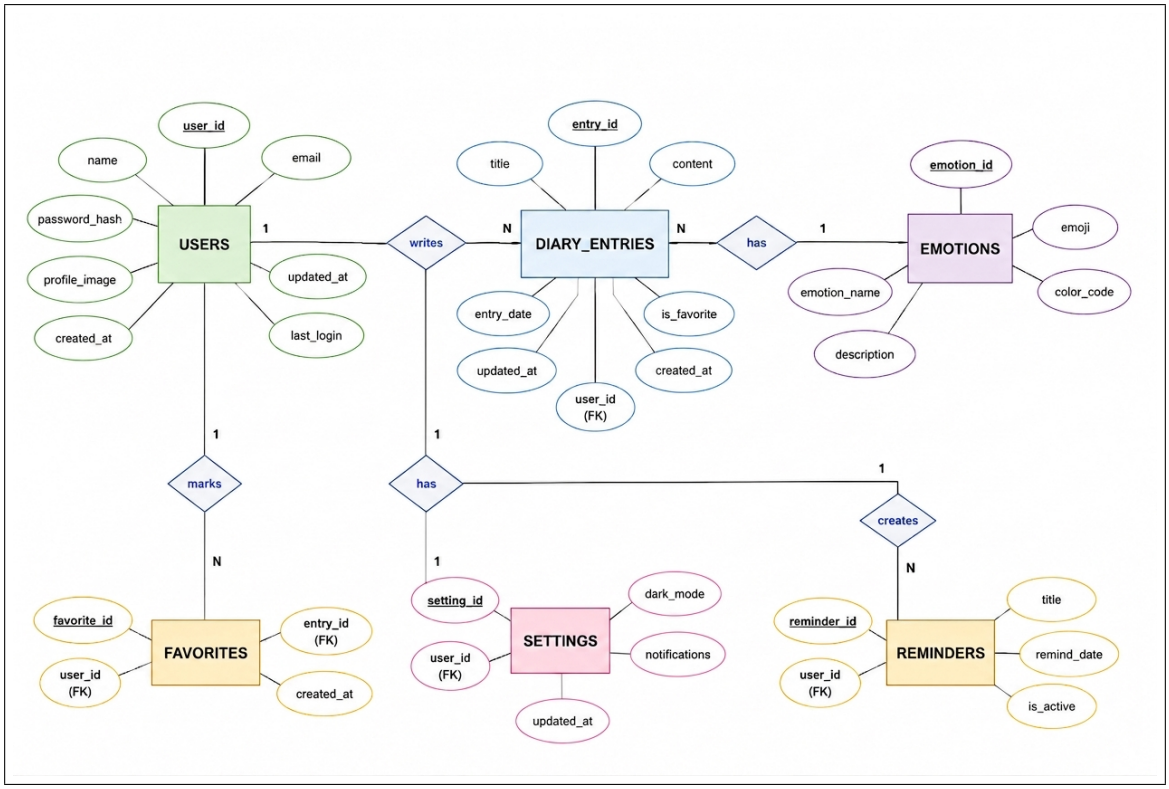


Figure 3.4: Entity-Relationship Diagram

This Entity-Relationship (ER) Diagram presents a comprehensive overview of the conceptual data architecture, entity attributes, and underlying structural connections governing the Emotional Diary Application. The design establishes a robust data modeling foundation aimed at ensuring logical data independence, high normalization, and strict enforcement of referential integrity constraints across all operational modules.

At the core of the architecture lies the USERS entity, which acts as the primary identity anchor within the system. To support personalized workspace management, the USERS entity maintains a 1-to-N (one-to-many) relationship with several dependent entities, including DIARY ENTRIES, FAVORITES, SETTINGS, and REMINDERS. This structural hierarchy guarantees that all journal logs, saved records, user preferences, and notification schedules remain strictly partitioned per individual user account.

Each record within DIARY ENTRIES is linked to the EMOTIONS entity to accurately assign primary emotional classifications and contextual metadata (such as emoji markers and visual color codes) to written logs. Furthermore, specific foreign key associations (such as userId, entryId, emotionId, and settingId) map relational pathways across secondary functions like favorite entry tracking, configurable user interface parameters, and automated reminder alerts, ensuring seamless data retrieval and structural consistency throughout the cloud backend.

3.3.6 Database Schema Diagram

The system uses a user-specific Firestore structure where authenticated users have their own diary collection. Each diary document may contain information such as title, content, mood/emotion, and relevant diary metadata.

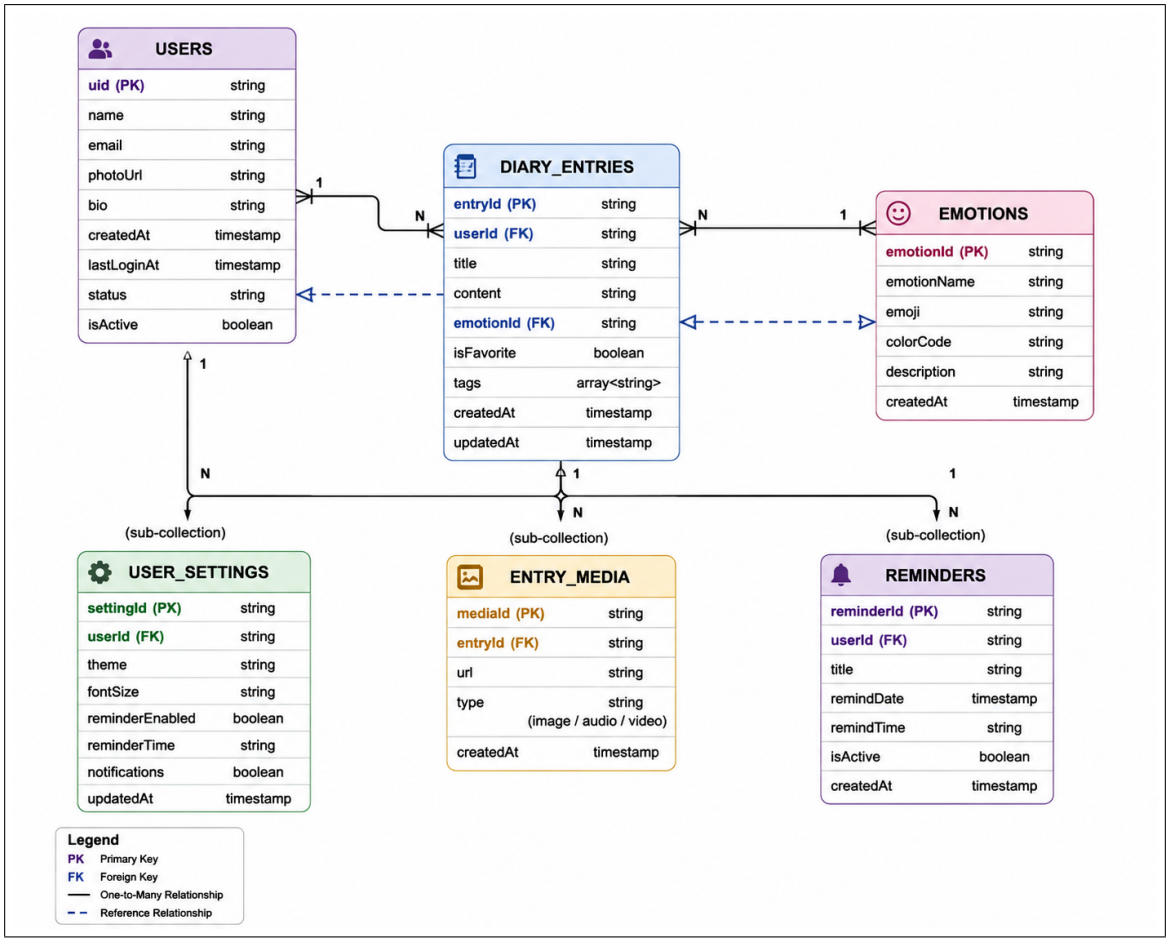


Figure 3.5: Database Schema Structure

This Database Schema diagram defines the NoSQL or Firestore-style structural design and relationships governing the Emotional Diary Application. The central USERS document maintains a one-to-many relationship with DIARY ENTRIES via the foreign key userId, as well as sub-collections including USER SETTINGS and REMINDERS to isolate user-specific configurations and scheduled notifications. Each diary entry in DIARY ENTRIES references the EMOTIONS collection through emotionId to map emotional context, while hosting its own sub-collection, ENTRY MEDIA, to support multi-modal attachments like image, audio, or video. Primary Keys and Foreign Keys across all schemas maintain data consistency, data typing, and structural integrity across the cloud database.

3.3.7 System Flowchart

The overall system flow begins with user authentication. After successful login, the user reaches the Home screen and can navigate to Diary or Profile sections. When a new diary is created, the system analyzes its emotional content, stores the result, and updates the emotion summary.

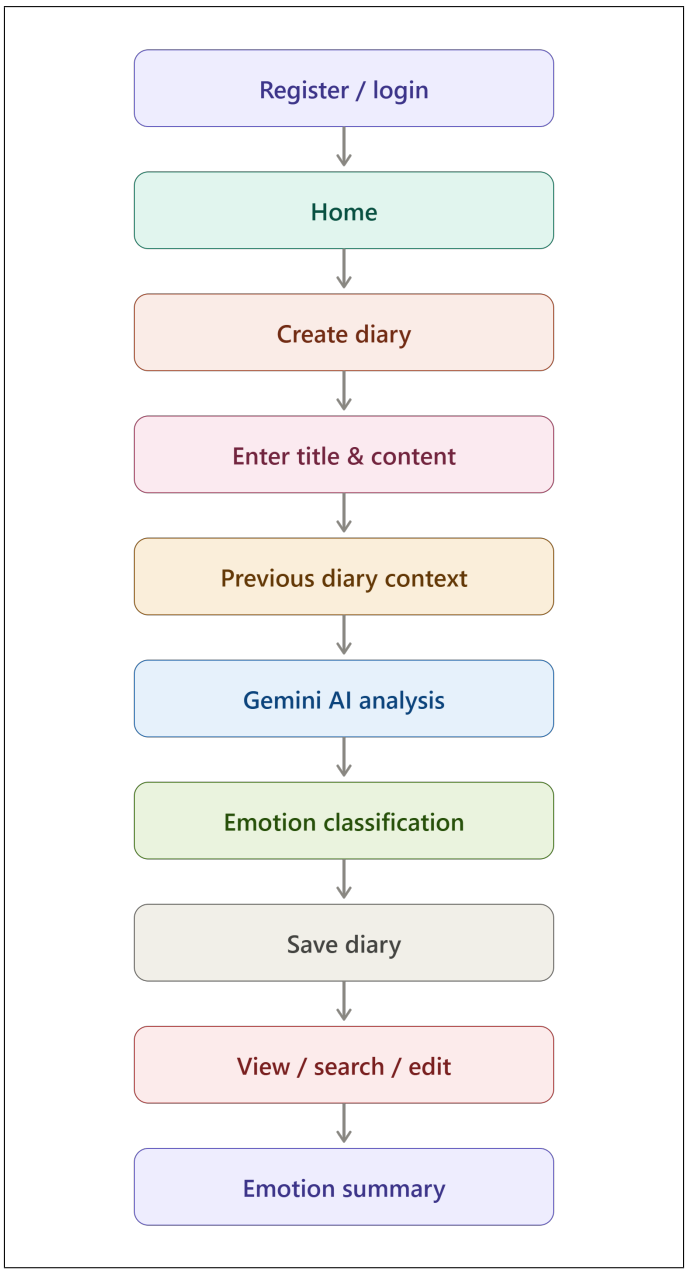


Figure 3.6: System Flowchart of Emotional Diary

This Flowchart Diagram illustrates the step-by-step user journey and operational logic within the Emotional Diary Application. The execution flow begins at Registration and Login, guiding the user to the Home dashboard to select Create diary. After inputting text under Enter title and content, the system pulls Previous diary context to process through Gemini AI analysis for automated Emotion classification. Once processed, the user completes the flow via Save diary, enabling downstream actions including View, Search, and Edit operations alongside generating an aggregate Emotion summary.

In summary, this chapter presented the methodology used to develop the Emotional Diary application. The feasibility study established the practicality of the project, while requirement analysis identified the functional and non-functional requirements. The system design described the architecture, database structure, user interactions, data flow, and overall system process.

Chapter 4

4 Implementation and Testing

The implementation phase transformed the proposed design into a functional Android application. The application was developed using Android Studio and Java. Different screens were implemented using XML layouts, while Firebase services were integrated for authentication and cloud data storage.
Testing was performed throughout the development process to identify functional errors, UI problems, database issues, authentication problems, and AI emotion classification issues.

4.1 System Setup

The development environment consisted of a computer with Android Studio, Java Development Kit, Android SDK, and an internet connection. The application was tested using Android devices and/or Android emulators.
Firebase services were configured for the application. Firebase Authentication was used to manage user accounts, while Cloud Firestore was configured to store user-specific diary information.
The Firebase AI environment was also configured for AI-assisted emotion analysis.

4.2 System Modules

The major modules implemented in the application are described below.

1. User Registration Module

The registration module allows a new user to create an account. The user provides the required account information and the application communicates with Firebase Authentication to create the account.

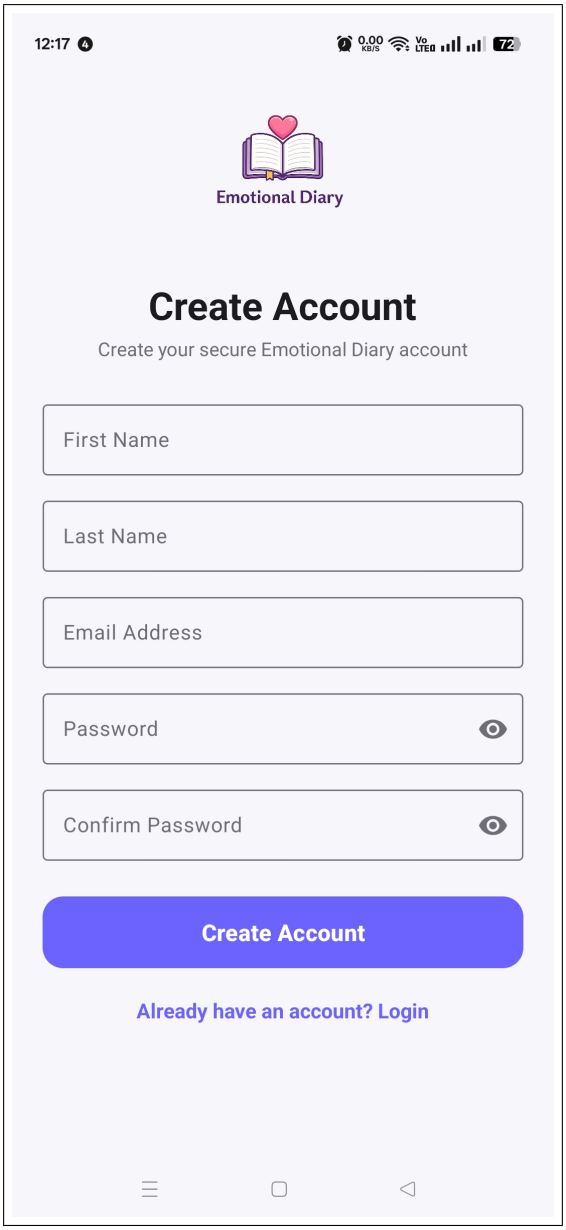


Figure 4.1: User Registration Interface

2. Login Module

The login module allows registered users to access their accounts. Firebase Authentication verifies the provided credentials. Successful authentication allows the user to access the main application interface.

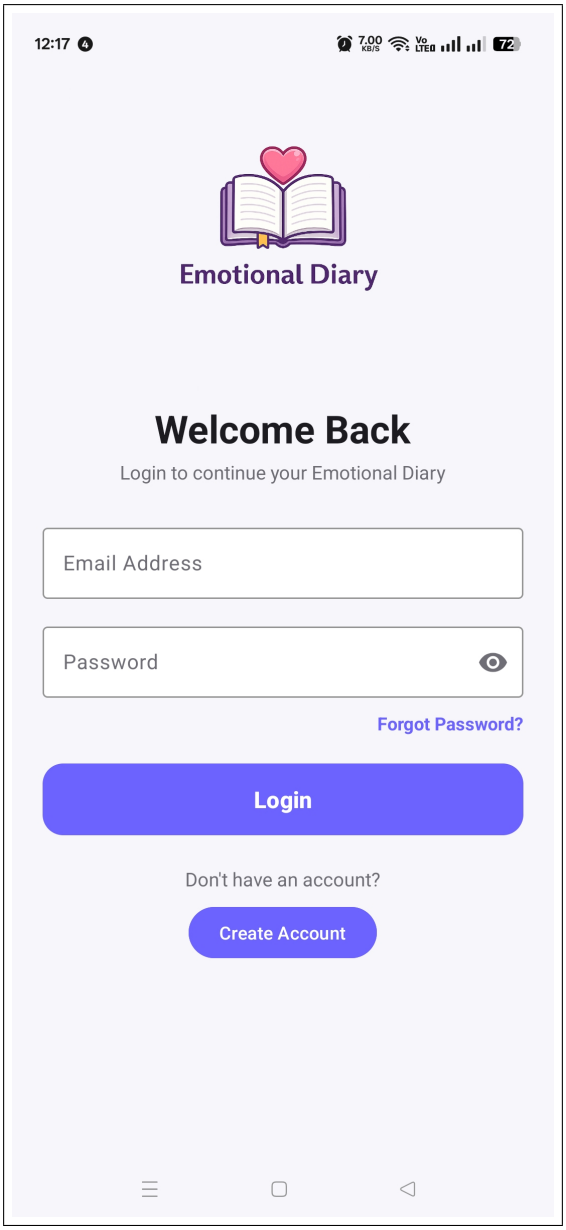


Figure 4.2: User Authentication / Login Interface

3. Home Module

The Home screen provides an overview of the user's diary activity. It contains a personalized greeting, a New Diary button, and an emotion summary section.
The emotion summary is presented visually through a pie chart. The chart represents the distribution of Happy, Excited, Calm, Neutral, Sad, and Angry diary entries.

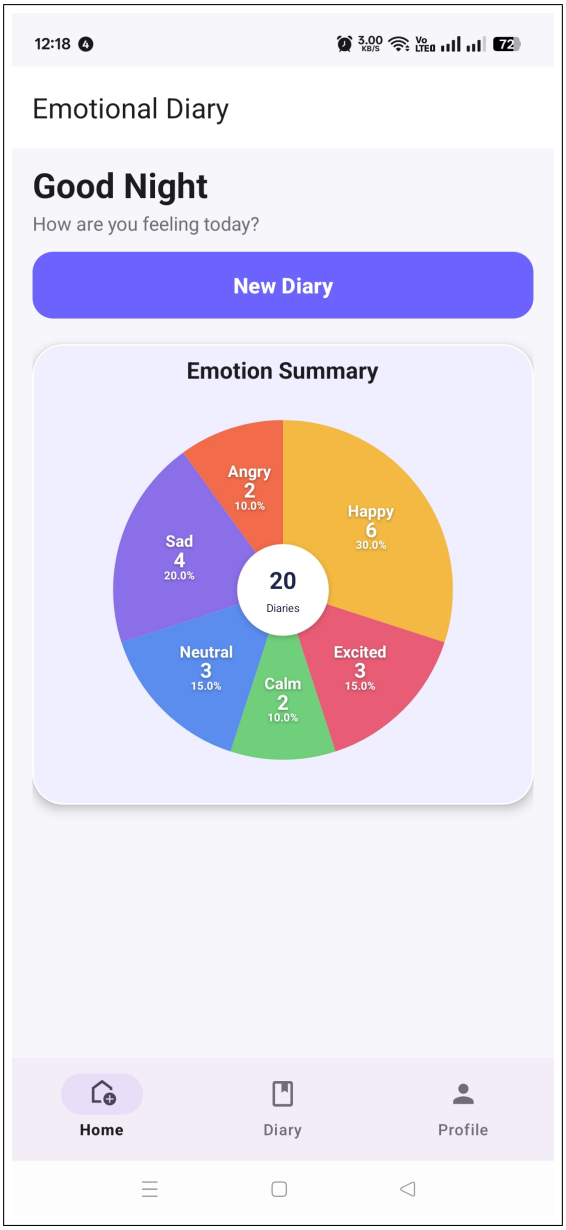


Figure 4.3: Home Screen Interface with Emotion Overview

4. Diary Management Module

The Diary module allows users to manage their personal diary entries. Users can create, view, edit, and delete diary entries.
A search field is also provided so that users can find diary entries more conveniently.

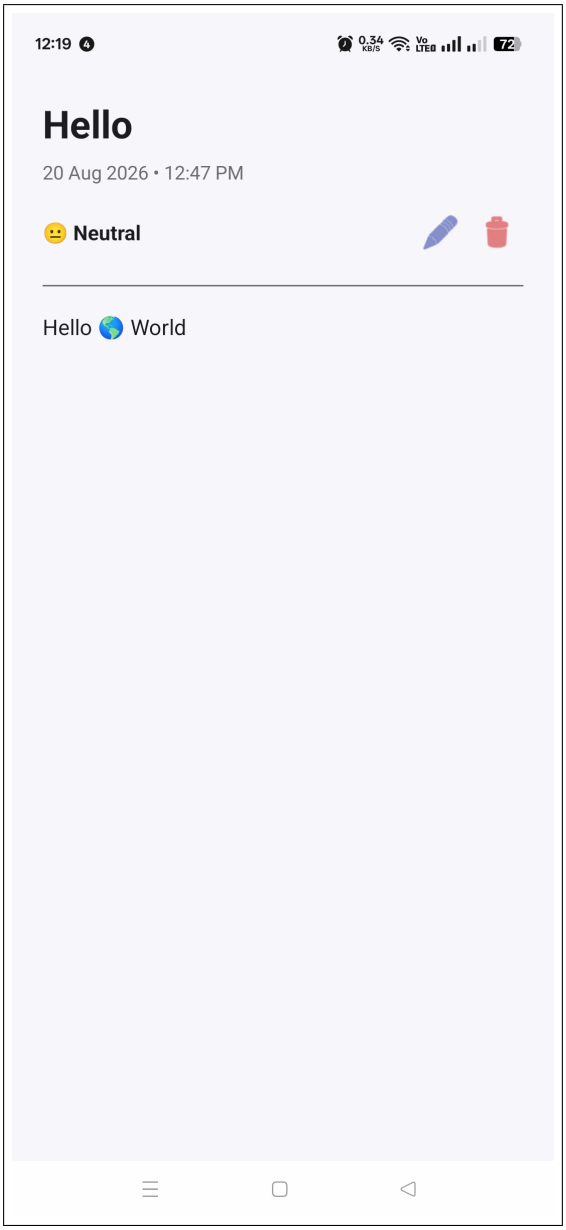


Figure 4.4: Diary Management and Search Interface

5. AI Emotion Detection Module

The AI emotion detection module is one of the most important components of the system. When a user writes a new diary entry, the system analyzes the title and content together. The AI system considers the overall meaning, emotional tone, context, events, feelings, and intentions expressed in the diary. The new diary is treated as the strongest evidence. Previous diary information is used only as supporting context. This allows the system to understand the user's personal writing style and emotional patterns without simply copying the emotion of previous entries.
The system restricts the final classification to six predefined categories:

- Happy
- Excited

- Calm
- Neutral
- Sad
- Angry

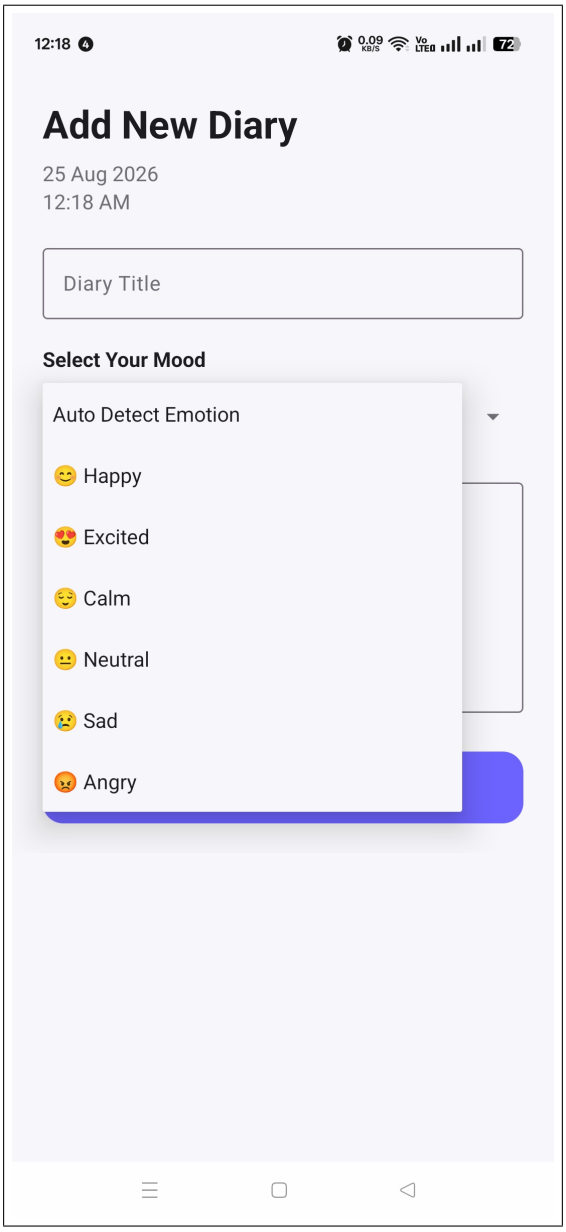


Figure 4.5: AI Emotion Detection Analysis Flow

6. Emotion Summary Module

After diary entries are stored, the system counts the number of entries associated with each emotion. These counts are passed to the custom emotion pie chart component. The visual summary helps users understand the general emotional distribution of their diary records.



Figure 4.6: Emotion summary pie chart

7. Profile Module

The Profile screen displays the authenticated user's name and email address. It also displays the first letter of the user's name inside a circular avatar.
The profile module provides access to:

- Edit Profile
- Change Password
- Settings
- Logout

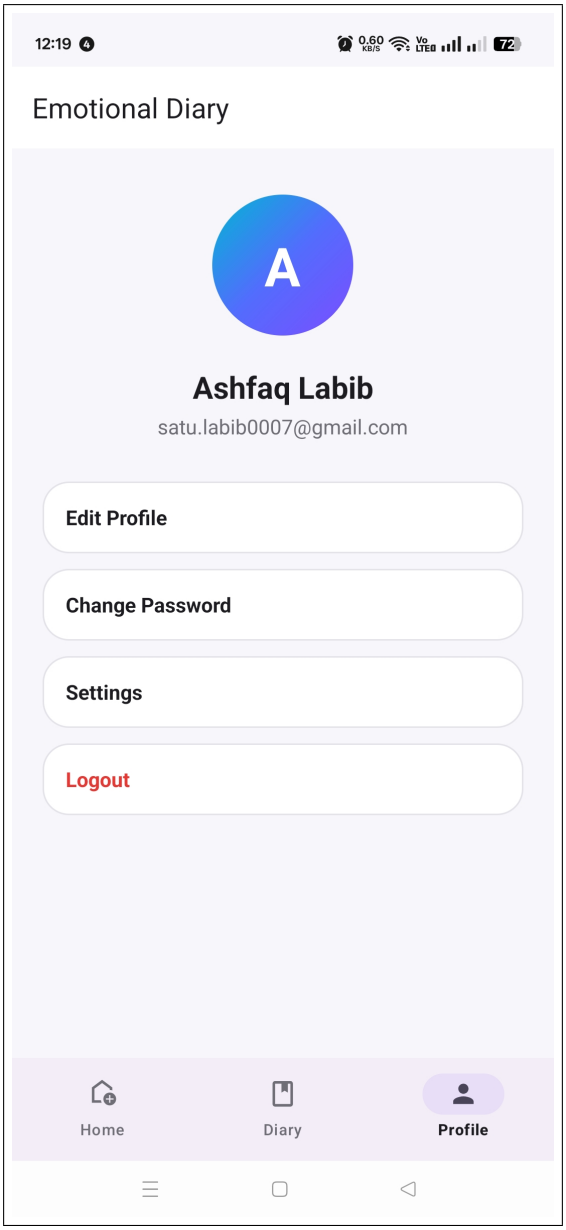


Figure 4.7: User Profile Interface

8. Settings Module

The Settings module provides application-level preferences. Theme customization is one of the important usability features of the application. Both light and dark themes were designed and adjusted to maintain readable text and appropriate contrast.

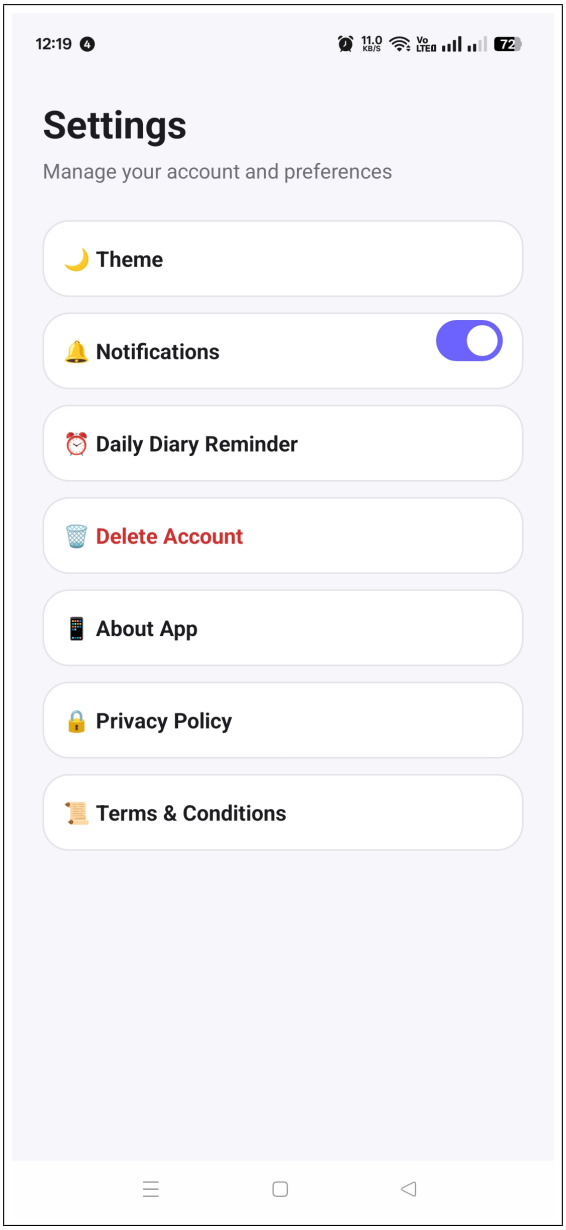


Figure 4.8: Settings Interface

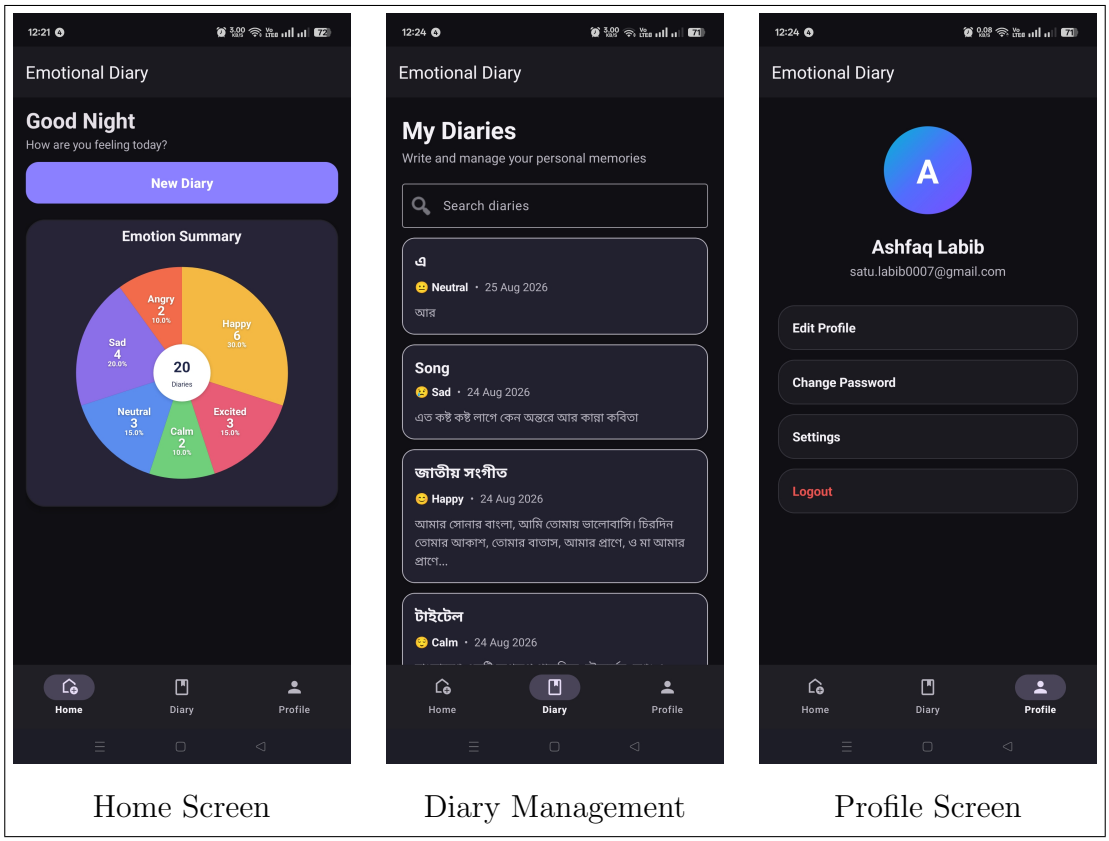
4.3 Prototype Design

The prototype of Emotional Diary was designed around a simple navigation structure. The main application contains three primary sections:

- Home
- Diary
- Profile

The Home screen provides an overview and emotion summary. The Diary screen focuses on diary management and searching. The Profile screen provides account and application management options.

The interface uses Material Design components and rounded cards and buttons to provide a modern and consistent visual appearance. The application also supports light and dark themes.



Home Screen | Diary Management | Profile Screen

Figure 4.9: Prototype design of the Emotional Diary application

4.4 System Requirements for End-User

The end-user requires the following:

- **Hardware Requirements:** An Android smartphone or compatible Android device.
- **Operating System:** Supported Android operating system version.
- **Storage Requirements:** Sufficient storage space for application installation.
- **Network Requirements:** Active internet connectivity for cloud-based authentication, database synchronization, and AI-based emotion analysis.
- **Account Requirements:** A valid email address for account registration and authentication.

4.5 Testing & Quality Assurance

Testing was performed to ensure that the application works correctly under normal operating conditions. Functional testing was performed for major application features.

Table 1: Functional Testing Summary

Test Module	**Expected Result**	**Status**
Registration	New account is created successfully	Passed
Login	Valid user can enter the application	Passed
Logout	User is returned to login screen	Passed
Create Diary	New diary is stored successfully	Passed
View Diary	Existing diary can be viewed	Passed
Edit Diary	Selected diary can be updated	Passed
Delete Diary	Selected diary can be removed	Passed
Search Diary	Matching diary entries are displayed	Passed
Emotion Detection	Dominant emotion is assigned	Passed
Emotion Summary	Emotion distribution is displayed	Passed
Edit Profile	Profile information can be updated	Passed
Change Password	Password can be changed	Passed
Dark Theme	UI remains readable in dark mode	Passed
Light Theme	UI remains readable in light mode	Passed

Quality assurance also included repeated testing after major code changes. Particular attention was given to Firebase operations, authentication states, diary data retrieval, emotion analysis results, theme consistency, and UI readability.

During development, Android resource linking and theme-related problems were identified and corrected. Duplicate color resources and incorrect theme attributes were also resolved to ensure successful application builds.

In summary, this chapter described the implementation of the Emotional Diary application and the testing activities performed to validate its major features. The completed prototype provides authentication, diary management, AI-based emotion analysis, emotion visualization, profile management, and theme support.

Chapter 5

5 Standards, Constraints, and Milestones

The development of a personal diary application involves considerations related to sustainability, social impact, ethics, technical constraints, and project planning. This chapter discusses these aspects of Emotional Diary.

5.1 Sustainability Standards

The application is designed as a software-based solution, reducing the need for paper-based diary records. Digital storage can reduce physical paper consumption and allows users to maintain their records electronically.
The use of cloud-based services also provides centralized data management. However, responsible management of cloud resources is necessary to avoid unnecessary data storage and processing.
The modular software structure also supports maintainability and future updates. New features can be added without completely redesigning the existing application.

5.2 Impacts on Society

Emotional Diary may have several positive social impacts. It provides users with a convenient way to record their personal experiences and reflect on their emotional states. The emotion summary can help users observe patterns in their own diary records. However, the system is intended as a journaling and reflection tool rather than a medical or psychological diagnostic system.
The project also demonstrates the practical use of artificial intelligence in everyday software applications and provides an example of how AI can be combined with mobile application development.

5.3 Ethics

Ethical considerations are particularly important because diary entries may contain highly personal information.
The system should handle personal information responsibly and provide appropriate authentication and access control. A user's diary information should not be intentionally exposed to other users.
The AI-based emotion classification should also be understood as an automated interpretation rather than an absolute statement about a user's mental or emotional condition. The application uses predefined emotion categories and attempts to select the dominant emotion based on the written content.

The system should not be presented as a replacement for professional medical, psychological, or mental-health assessment.

5.4 Challenges

Several challenges were encountered during the development of the application.

- **Firebase Authentication:** Integrating Firebase Authentication with the Android application.
- **Firestore Data Management:** Structuring user-specific diary data in Firestore.
- **Asynchronous Operations:** Managing asynchronous Firebase operations.
- **AI Integration:** Integrating AI-based emotion detection into the diary workflow.
- **Prompt Engineering:** Designing a reliable prompt for consistent emotion classification.
- **Previous Diary Context:** Ensuring that previous diary context supports rather than dominates the new diary analysis.
- **AI Response Handling:** Handling AI responses that may contain additional formatting or unexpected text.
- **Theme Consistency:** Maintaining UI readability in dark mode.
- **Resource Debugging:** Resolving Android resource linking and duplicate resource errors.
- **Testing:** Testing multiple screens and navigation flows.

5.5 Constraints

The major constraints of the project include:

Design Constraints

The application must provide a simple interface suitable for mobile screens. The number of visible components must therefore be balanced with readability and ease of navigation.

Technical Constraints

The application depends on Android, Firebase services, internet connectivity, and AI services. Temporary network problems may affect cloud operations and AI-based emotion analysis.

AI Constraints

AI-based emotion classification may not always perfectly represent the user's actual emotional state. Human language can contain sarcasm, mixed emotions, ambiguous expressions, and context that is difficult to interpret automatically.

Data Constraints

The quality of emotion analysis depends on the information provided in the diary. Very short or highly ambiguous diary entries may result in a Neutral classification.

Budget Constraints

As an academic project, development was performed using available hardware, software, and cloud resources. The project was designed to avoid requiring expensive dedicated infrastructure.

5.6 Timeline and Gantt Chart

The project was completed through several development stages. A representative project timeline is shown below.

Table 1: Project Development Timeline

Activity	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**
Requirement Analysis	✓	✓				
UI Design		✓	✓			
Authentication			✓	✓		
Diary Management				✓	✓	
AI Emotion Detection				✓	✓	
Emotion Summary					✓	
Profile and Settings					✓	
Testing and Debugging					✓	✓
Final UI Polishing						✓
Documentation						✓

The development timeline was iterative, meaning that testing and debugging were performed alongside implementation rather than only at the end of the project.
In summary, this chapter discussed the sustainability, social impact, ethical considerations, challenges, constraints, and development milestones of Emotional Diary. These considerations helped ensure that the application was developed as a practical and responsible academic software project.

Chapter 6

6 Conclusion

Emotional Diary was developed as a secure and intelligent personal diary mobile application that combines conventional diary management with AI-assisted emotion analysis. The application provides users with a structured environment for creating, viewing, editing, deleting, and searching personal diary entries.

The project successfully integrates Android application development with Firebase Authentication and Cloud Firestore. Authentication allows users to maintain individual accounts, while Firestore provides cloud-based storage for diary records.

The most significant feature of the application is its AI-based emotion detection system. The system analyzes a newly written diary by considering its title, content, context, emotional tone, events, feelings, and overall meaning. It also uses previous diary context as supporting information to understand the user's personal writing patterns.

The application restricts emotion classification to six categories: Happy, Excited, Calm, Neutral, Sad, and Angry. The resulting emotion data is then used to create an emotion summary through a pie chart.

In addition to the core diary functionality, the application provides profile management, password management, settings, logout functionality, and light/dark theme support. The final implementation demonstrates the practical integration of mobile development, cloud computing, database management, artificial intelligence, and user interface design.

6.1 Limitations

Although the application provides the required functionality, several limitations remain.

- **Classification Accuracy Limit:** AI-based emotion classification may not always perfectly represent the user's actual emotional state.
- **Fixed Emotion Categories:** The system currently uses a predefined set of six emotion categories.
- **Connectivity Dependency:** AI-based analysis depends on internet connectivity and availability of the required AI service.
- **Input Length Sensitivity:** Very short or ambiguous diary entries may be difficult to classify accurately.
- **Non-Psychological Nature:** The emotion summary is based on the recorded diary entries and does not represent a professional psychological assessment.
- **Text-Only Scope:** The current application focuses primarily on text-based diary entries and does not provide advanced multimedia journaling features.

6.2 Future Works and Direction

Several improvements can be considered for future versions of Emotional Diary.

- **Advanced Emotion Classification:** Introduce a larger and more detailed emotion classification system.
- **Long-Term Emotion Analysis:** Provide weekly, monthly, and yearly emotional trend analysis.
- **Advanced Dashboard:** Add graphical dashboards for more detailed emotional insights.
- **Diary Reminders:** Introduce optional reminders for daily journaling.
- **Multimedia Diary:** Add multimedia diary support including images, audio, and other attachments.
- **Offline Support:** Improve offline functionality and synchronize data when internet connectivity becomes available.
- **Biometric Protection:** Add stronger privacy mechanisms such as application-level biometric protection where appropriate.
- **Personalized AI Insights:** Provide more personalized AI-assisted diary insights while maintaining user privacy.
- **Advanced Emotion Visualization:** Improve the visual analytics section with additional charts and emotion trends.
- **Personalization:** Provide more advanced personalization and customization options.

Overall, Emotional Diary demonstrates how a modern mobile application can combine secure cloud-based diary management with artificial intelligence to create a more meaningful and personalized journaling experience. The completed system provides a strong foundation for further development and future enhancement.

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